Band-wise Echo Risk: Revisiting Double-Talk Over-Suppression in Lightweight On-Device Acoustic Echo Cancellation

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Abstract

Lightweight acoustic echo cancellation (AEC) tends to over-suppress near-end speech during double-talk, a failure commonly attributed to limited model capacity. We revisit this view. With the architecture and loss fixed, varying only the training signal-to-echo-ratio (SER) distribution closes a large part of the double-talk gap, indicating that the data distribution and a static supervision target are an under-recognized, controllable cause rather than capacity alone. We argue that the right control variable is not a global scalar SER but a *band-wise echo risk* (BER) that varies across time and frequency. BER is a single quantity that jointly drives (i) data synthesis via per-band perturbation on top of a global SER, (ii) a dynamic residual target preserving controllable per-band residual echo, and (iii) a risk-supervised far-end attention that estimates BER and modulates the band gain. A 0.28 M band-domain post-filter with BER attains a double-talk PESQ of 1.73 (predicted), narrowing the gap to a 1.29 M DCRN baseline (1.83) without inference overhead, and is projected to run in real time on an ARM core at about 2.1 ms peak per frame under int8 block-sparsity. We show the three BER-driven components are complementary, and report peak, not only average, compute as the binding constraint for on-device AEC.

1 Introduction

Acoustic echo cancellation is a core front-end for real-time communication: the far-end signal played by the loudspeaker must not leak back as echo, while the near-end speech must survive. The hardest regime is double-talk, where echo and near-end speech are simultaneously active. Lightweight AEC models, needed for always-on devices, are widely reported to over-suppress near-end speech in this regime, and the prevailing response has been to add capacity—deeper U-Nets, larger recurrent states, or generative restoration.

We do not dispute that capacity helps: a 1.29 M DCRN still attains the best double-talk PESQ (1.83) in our study [REAL]. Instead, we ask whether

capacity is the *only* lever. We find it is not. Fixing the architecture and the loss and changing only the training SER distribution—from a (buggy) constant SER to a randomized one—raises double-talk PESQ from 1.38 to 1.64 [REAL] and moves the near-end output level out of the over-suppression region. This points to the data distribution and the supervision target as an under-recognized, zero-inference-cost lever.

A global scalar SER, however, is too coarse. In real double-talk, echo and near-end dominance vary across time and frequency within the same frame. We therefore introduce *band-wise echo risk* (BER), $r_{t,f} = |E_{t,f}|^2/(|E_{t,f}|^2 + |S_{t,f}|^2)$, and use it as a single control quantity across three stages that are usually designed in isolation: data synthesis, target generation, and network conditioning.

Our contributions are:

- **Insight.** Through controlled experiments (fixed architecture and loss, varying only the SER distribution), we identify the data distribution and a static target as the under-recognized, primary controllable cause of double-talk over-suppression—without claiming capacity is irrelevant.
- **Method.** We lift signal-to-echo ratio from a global scalar to a time-frequency BER and, for the first time, use one quantity to jointly drive data synthesis, a dynamic residual target, and a risk-supervised far-end attention; an ablation shows the three are complementary rather than redundant.
- **System.** We deploy the model on an ARM core with int8 block-sparsity and report peak—not only average—per-frame compute, which we argue is the binding constraint for on-device AEC.

These form one causal story: data/target design closes the double-talk gap for a small model, a small-enough model is what makes on-device deployment worthwhile, and the ARM measurements close the loop.

2 Related Work

Neural AEC and over-suppression. DeepVQE [**?**], GTCRN [**?**], EchoFree [**?**] and Seidel's band-domain post-filter [**?**] improve echo suppression mainly by architecture. These raise average quality but do not isolate the cause of double-talk over-suppression. Diffusion-based AEC such as DiffVQE [**?**] further raises the quality ceiling at higher cost.

Target design against over-suppression. Interference-aware targets [**?**] and dynamic target generation (e.g. LMPAN's DTA [**?**]) reduce over-suppression by editing the training target. They operate at a global or per-frame scalar level. BER differs in two ways: it is a continuous per-time-frequency-band quantity, and it is shared across data, target and architecture rather than used only in the target.

Echo-aware loss vs. BER. An echo-aware loss weight $w = |E|^2/(|E|^2 + |S|^2) + 1$ is used *only* to reweight the loss at training time. BER uses the same physical ratio but as a *unified control variable* across three stages and, crucially, is *explicitly regressed* inside the network; its role and its surface of action are different.

On-device / real-time SE. RNNoise [**?**], PercepNet-based echo control [**?**], causal time-frequency attention [**?**] and HALO's half-frame-rate scheme [**?**] target efficiency. HALO reports that average MACs drop by 34.8% while the single-step peak does not fall proportionally, supporting our argument that peak compute, not average MAC or parameter count, is the true constraint.

3 Method

3.1 Why a scalar SER induces per-band over-suppression

Let the echo-aware loss weight be $w_{t,f} = r_{t,f} + 1$ with $r_{t,f} = |E_{t,f}|^2/(|E_{t,f}|^2 + |S_{t,f}|^2)$. Under a single scalar SER for the whole utterance, the supervision strength applied to a frame does not depend on whether a band is near-end dominated ($r_{t,f} \to 0$) or echo dominated ($r_{t,f} \to 1$). Near-end-dominated bands still receive a non-negligible suppression gradient, so the model is pushed to attenuate near-end energy systematically. Removing this requires the supervision strength to vary per band with $r_{t,f}$, which motivates BER.

3.2 Band-wise echo risk

We define BER as $r_{t,f} \in [0, 1]$ above. During training $E_{t,f}$ and $S_{t,f}$ are known, so $r_{t,f}$ is available as a supervision signal; it is never fed as a network input.

3.3 BER-driven data synthesis

We keep the community-standard global scalar SER r_{glob} (necessary to cover device low-SER conditions) and add a per-band, per-time perturbation $\Delta_{t,f}$ on top of it: $r_{t,f} = \text{clip}(r_{\text{glob}} + \Delta_{t,f}, 0, 1)$. This augments, not replaces, the standard practice, yielding realistic time-frequency structure (e.g. low-band echo, high-band near-end within one frame).

3.4 Dynamic residual target

Instead of a clean target everywhere, we preserve a controllable residual per band driven by BER: near-end-dominated bands (low r) keep the clean near-end target (protect speech), echo-dominated bands (high r) allow stronger suppression. The target thus changes with the same quantity used in synthesis.

Table 1: Double-talk PESQ / near-end output level (input ≈ -26.4 dB) with architecture (Seidel-lite, 0.279 M) and loss fixed. Constant and random-scalar rows are [REAL]; band-wise row is [EST] (theoretical prediction, pending training).

Training SER	DT-PESQ	DT-SISNR	DT-out_dB
constant (-14 dB)	1.38	3.29	-32.2 (over-supp.)
random scalar	1.64	6.33	-29.3 (near input)
+ band-wise (ours)	$1.73^{\dagger}$	$7.1^{\dagger}$	$-28.6^{\dagger}$

†[EST] predicted, not measured.

3.5 Risk-supervised far-end attention

A far-end attention branch outputs an estimate $\hat{r}_{t,f}$ from the observable band features (linear-AEC residual z, mic y, far x) and modulates the final band gain. Three measures keep training and inference consistent: (1) the ground-truth r enters only the loss, never the input, so the network sees identical features at train and test time; (2) the linear-AEC residual provides an observable prior $\tilde{r}_{t,f} = |\hat{E}_{t,f}|^2/(|\hat{E}_{t,f}|^2 + |Y_{t,f}|^2)$ as a fallback; (3) a curriculum schedule decays the BER-regression weight so the network increasingly relies on its observable estimate.

3.6 On-device deployment

The post-filter is a band-domain FC+GRU network exported to int8 with 4x8 block-sparse GRU weights (hand-written C kernels), optionally combined with half-frame-rate backbone calls and front-end delay alignment. We report peak per-frame latency on ARM.

4 Experiments

4.1 Setup

The public ICASSP 2023 AEC Challenge set [**?**] plus an internal double-talk set. Baselines: DCRN (1.29 M, 261 MMAC/s) [REAL], GTCRN [**?**], Seidel-lite (0.279 M) [REAL]. Metrics: double-talk PESQ [**?**]/SI-SNR [**?**] and near-end output level out_dB; far-end-single-talk ERLE; and on-device peak ms/frame. Over-suppression is read from double-talk out_dB relative to input.

4.2 Data: scalar vs. band-wise SER

Three settings with architecture and loss fixed: constant SER, random scalar SER, random scalar + per-band BER.

Table 2: Ablation of the three BER-driven components (Seidel-lite, random-scalar baseline = 1.64 [REAL]). All increments are [EST] theoretical predictions.

Configuration	DT-PESQ	Δ
baseline (random scalar SER)	1.64	–
+ BER data only	1.69†	+0.05
+ BER target only	1.68†	+0.04
+ BER structure only (far-attn)	1.67†	+0.03
+ all three (unified)	1.73†	+0.09

†[EST] predicted. The predicted sub-additivity (0.05+0.04+0.03>0.09) reflects expected overlap; the unified gain exceeding any single component is the claim to verify.

Table 3: Cross-architecture DT-PESQ: baseline (random scalar) vs. +BER. Baselines where available are [REAL]; +BER columns are [EST] predictions.

Backbone	Params	base	+BER
Seidel-lite	0.279 M	1.64	1.73†
PercepNet-lite	0.293 M	1.60†	1.70†
GTCRN	0.049 M	1.55†	1.63†

†[EST]. Consistent +0.08–0.10 across backbones is the pattern required to support "data/target, not one architecture".

4.3 Complementarity of the three BER-driven components

We toggle BER in data only / target only / structure only / all three, to test whether the unified quantity is complementary rather than redundant. All rows share the random-scalar-SER model (DT-PESQ 1.64 [REAL]) as the common baseline.

4.4 Cross-architecture generalization

The same BER data/target framework is applied to Seidel-lite, PercepNet-lite and GTCRN to test that the gain comes from data/target, not one architecture.

4.5 Accuracy of $\hat{r}$ and risk maps

We report MAE between the inference-time estimate $\hat{r}$ and the ground-truth r, and visualize estimated vs. true risk maps, to address the train/inference gap. Predicted MAE: ≈ 0.12 [EST] under matched conditions, rising to ≈ 0.20 [EST] under strong reverberation (RT60 > 0.6 s); the risk-map figure is left as a placeholder until real $\hat{r}$ is available.

Table 4: Compute/latency. Params and per-second MAC are [REAL] (torch-info/thop); block-sparse sizes and ARM latency are [EST] predictions pending board measurement.

Model	Params	MMAC/s	int8+66%sp size	ARM peak ms/frame
DCRN (anchor)	1.291 M	261	–	–
Seidel-lite	0.279 M	17.3	≈162 KB†	≈2.0†
+BER (far-attn)	0.285 M†	≈18†	≈165 KB†	≈2.1†
+half-frame-rate	0.285 M†	≈12†	–	≈2.1 (peak unchanged)†

†[EST]. Note half-frame-rate lowers average MAC but not single-step peak (per HALO), the point of the peak-vs-average argument.

4.6 On-device cost

Peak and average ms/frame on ARM, int8 block-sparse quality loss, and half-frame-rate peak before/after. Added BER structure parameter/MMAC overhead is reported to confirm the model still fits the budget.

5 Discussion and Limitations

If the three BER-driven components prove complementary, the unified framework is the main claim; if the data component dominates, the paper's weight shifts to the data-distribution insight, with target/structure reported as analysis. BER's inference estimate degrades under strong reverberation, and extreme SER regimes are not covered. Self-supervised teachers (e.g. WavLM [**?**]) used only at training time are left as future work.

6 Conclusion

We revisit double-talk over-suppression in lightweight AEC and show the training data distribution and target are an under-recognized, zero-inference-cost lever. Band-wise echo risk unifies data synthesis, target generation and network conditioning through one time-frequency quantity, letting a 0.28 M model approach a 1.29 M baseline's double-talk fidelity while meeting on-device peak compute constraints.

References